

IACS UR E27 — Cyber Résilience of On-board Systems & Equipment

IACS Unified Requirement E27

MV Demo Carrier

AI-powered Analysis

ENTITY	MV Demo Carrier
STANDARD	IACS E27
REPORT TYPE	Internal Audit & Compliance Readiness Assessment
DATE	12 August 2026
PREPARED BY	Inside the Box
STATUS	CONFIDENTIAL

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PART A — EXECUTIVE REPORT

Part A is written for management and decision-makers: readiness, scope, risks, action plan and conclusion. Requirement-level evidence sits in Part B.

1 Executive Summary

Demo readiness assessment for QA rendering.



Readiness is a preparedness indicator derived from the recorded answers. It is not a conformity statement, certification or class decision.

COMPLIANCE DISTRIBUTION

VERDICT OVERVIEW



Applicability

Source: Deterministic Assessment Engine

Applicability	Applicable
CBS of E22 category III (safety-critical): UR E27 applies to this system; the required capabilities follow from the category and the target security level.	
Assessed CBS / supplier	NaviGuard ECDIS v4
Rôle	supplier
E22 system category	III
Target security level	SL 2

Management Attention Index

Source: Deterministic Assessment Engine

Management Attention Index	Critical (Critical 0 · High 1 · Medium 1 · Low 0)
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KEY DRIVERS

- %, 2 mandatory control gap(s)
- %, 1 high risk(s)
- %, Readiness 79%

Audit Readiness

Source: Deterministic Assessment Engine

Readiness (overall)	Developing · 54%
Documentation	Limited · 20%
2/10 controls backed by documented evidence (policy/procedure/document/audit report).	
Operational	Substantial · 79%
Effectiveness of implemented controls (weighted compliance score).	
Governance	Strong · 80%
No dedicated governance category — overall score used.	
Evidence	Developing · 38%
10/10 controls with evidence, weighted by evidence strength.	

Deterministic Root Causes

Source: Deterministic Assessment Engine — derived from the findings, not AI-generated

The 2 open findings concentrate in 2 root-cause themes. Resolving these themes addresses the majority of individual gaps.

RC1 Insufficient Protect

Affected Controls	E27-PR4
Below Conformity	1 (1 gap, 0 partial)

BUSINESS IMPACT

Increased likelihood of non-conformities and findings in this domain.

RC2 Insufficient Recover

Affected Controls	E27-RC1
Below Conformity	1 (1 gap, 0 partial)

BUSINESS IMPACT

Slower detection of and recovery from disruptions and incidents.

2 Scope, Applicability and Claims

The following section documents the intake data captured for the assessed entity. These form the basis of the rule-based assessment.

Applicability

Applicability	Applicable
CBS of E22 category III (safety-critical): UR E27 applies to this system; the required capabilities follow from the category and the target security level.	

Scope in detail

Assessed CBS / supplier	NaviGuard ECDIS v4
Rôle	supplier
E22 system category	III

The category determines which E27 requirements are mandatory and which are conditional.

Target security level	SL 2
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What this report claims

- %, Readiness of the individual CBS against the assessed UR E27 capabilities, based on the recorded answers.
- %, Prioritised capability and documentation gaps with evidence needs.

What this report does not claim

- %, No type approval, class certificate or security-level confirmation by a classification society.
- %, No statement about the vessel level under UR E26 — that requires a separate E26 assessment.
- %, No statement about IEC 62443 certification, IMO 2021/ISM, NIS2 or ISPS; references are contextual.
- %, Evidence was not independently verified unless explicitly marked as verified.

Recorded intake data

ENTITY
MV Demo Carrier

SYSTEM / SUPPLIER

NaviGuard ECDIS v4

RÔLE

Manufacturer / supplier

TARGET SECURITY LEVEL (IEC 62443)

SL 2 — against intentional, simple attacks

DESCRIPTION OF THE SYSTEM, INTERFACES & INTENDED USE

Type-approved ECDIS unit running a hardened Linux variant. Ethernet and serial (NMEA) interfaces, role-based local accounts, signed software updates. Supplied to yards for newbuild integration; intended as a safety-critical Category III system at target SL 2.

SYSTEM CATEGORY BY IMPACT

Category III — safety-critical

ESTABLISHED PROCESSES & EVIDENCE

Secure development lifecycle (SDL), Test / type-approval reports available, Security / hardening manual, Vulnerability / patch process

KNOWN WEAKNESSES / OPEN POINTS

Audit logging exists but is not exported to a central collector. Configuration backup/restore is manual. SBOM not yet provided to integrators. NMEA serial inputs lack authentication and rely on segmentation.

SECURITY CAPABILITIES (UR E27 / IEC 62443)

Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration

3 Assessment Principles

The assessment follows an audit-safe, three-layer model: the compliance decision (pass / partial / gap) is derived strictly from the intake answers by deterministic rules. No finding is invented by the AI. Risks are derived deterministically from the gaps. The AI is used solely for the explanatory analysis layer.

How This Assessment Was Produced

- % Assessment responses were collected through structured intake.
- % Each requirement verdict (pass / partial / gap) was determined by deterministic rules over those answers.
- % Readiness percentages, risks, actions and the roadmap were computed from those verdicts.
- % Evidence was classified by type and verification level; unverified statements are marked as self-declared.

%, AI generated explanatory insights and hypotheses only; it never created, modified or overrode a verdict, risk or evidence record.

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4 Requirement-level Readiness Matrix

Source: Deterministic Assessment Engine

Complete, traceable overview of all assessed requirements with verdict.

ID	Article	Topic	Area	Verdict	Evidence	Action
E27-LC1	UR E27 §3	Secure development lifecycle	Development &	Pass	Documented	—
E27-LC2	UR E27 §3	Vulnerability, patch & updat	Development &	Pass	Declared	—
E27-ID1	UR E27 FR	Identification & authenticat	Identify	Pass	Declared	—
E27-PR1	UR E27 FR2	Use control & authorisation	Protect	Pass	Declared	—
E27-PR2	UR E27 FR3	System integrity & hardening	Protect	Pass	Declared	—
E27-PR3	UR E27 FR4	Data confidentiality (FR4)	Protect	Pass	Declared	—
E27-PR4	UR E27 FR5	Restricted data flow (FR5)	Protect	Gap	Declared	REC-01
E27-DE1	UR E27 FR6	Audit logging & timely respo	Detect	Pass	Verified	—
E27-RS1	UR E27 FR6	Timely response to security	Respond	Pass	Declared	—
E27-RC1	UR E27 FR7	Resource availability, robus	Recover	Gap	Declared	REC-02

Evidence column shows the verification level of the supporting evidence, not its quantity. Action column references the action in section 6.

Positions requiring attention

Every non-passing requirement in one line: what is missing and which action closes it. Full evidence detail is in Part B.

E27-PR4

UR E27 FR5 — Restricted data flow (FR5)

Identified Gap: Not evidenced: Restricted data flow (FR5).

Action: REC-01 · Critical · Due within 0–3 months

E27-RC1

UR E27 FR7 — Resource availability, robustness & recovery (FR7)

Identified Gap: Not evidenced: Resource availability / robustness (FR7).

Action: REC-02 · Critical · Due within 0–3 months

5 Risk Landscape

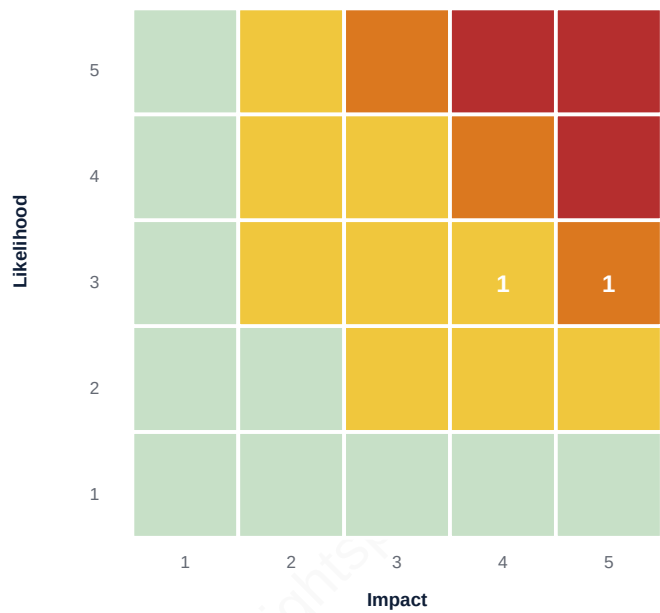
Source: Deterministic Risk Engine

The following risks were derived deterministically from the identified gaps (likelihood × impact).

RISK DISTRIBUTION



RISK HEATMAP



FAIL Efficiency in: Resource availability, robustness & recovery (FR7) (Impact 5 × Likelihood 3 = 15)
Business consequence: Slower detection of and recovery from disruptions and incidents.

PARTIAL Efficiency in: Restricted data flow (FR5) (Impact 4 × Likelihood 3 = 12)
Business consequence: Increased likelihood of non-conformities and findings in this domain.

6 Action Plan and Roadmap

Prioritised remediation plan, derived from the gaps and risks.

Each action states the deliverable, the acceptance criterion and how completion is verified, so it can be assigned and closed out.

REC-01 Implement / evidence: Restricted data flow (FR5).

Critical · Due within 0–3 months · E27-PR4

Deliverable	Implemented control plus documented evidence
Acceptance criterion	Requirement is implemented and the evidence has been reviewed.
Verification	Effectiveness test plus evidence review
Owner	Integrator

Direct regulatory & operational exposure

REC-02 Implement / evidence: Resource availability / robustness (FR7).

Critical · Due within 0–3 months · E27-RC1

Deliverable	Implemented control plus documented evidence
Acceptance criterion	Requirement is implemented and the evidence has been reviewed.
Verification	Effectiveness test plus evidence review
Owner	Manufacturer / ship

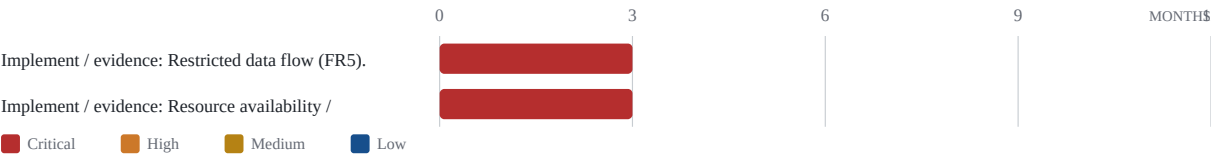
Direct regulatory & operational exposure

Remediation Roadmap

PHASE 0-3 MONTHS

- %, REC-01 · Critical — Implement / evidence: Restricted data flow (FR5).
- %, REC-02 · Critical — Implement / evidence: Resource availability / robustness (FR7).

Remediation Timeline



7 Conclusion

Demo readiness assessment for QA rendering.

Applicability	Applicable
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This report does not constitute a formal certification and does not replace assessment by a recognised authority. The compliance assessment is based on the information provided during intake.

WHAT THIS REPORT DOES NOT CLAIM

- %, No type approval, class certificate or security-level confirmation by a classification society.
- %, No statement about the vessel level under UR E26 — that requires a separate E26 assessment.
- %, No statement about IEC 62443 certification, IMO 2021/ISM, NIS2 or ISPS; references are contextual.
- %, Evidence was not independently verified unless explicitly marked as verified.

PART B — EVIDENCE PACK

Part B is the evidence pack for verifiers: requirement by requirement with answer, evidence, verification level and rule logic.

8 Requirement-level Evidence and Verification

Source: Deterministic Assessment Engine

Each requirement was assessed against the captured evidence using deterministic rules. Gaps and partial compliance are accompanied by concrete remediation recommendations.

8.1 E27-LC1 — Secure development lifecycle & evidence (UR E27 §3)

PASS

ASSESSMENT QUESTION

Has the organization implemented and evidenced "Secure development lifecycle & evidence" as required by UR E27 §3?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1)

Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration

Established processes & evidence

Secure development lifecycle (SDL), Test / type-approval reports available, Security / hardening manual, Vulnerability / patch process

EVIDENCE

Evidenced: Security documentation for integration, Secure development lifecycle (SDL), Test / type-approval reports available.

Verification level

Documented

Needed to verify: Confirm effectiveness through testing or independent review.

RULE LOGIC AND RESULT

% , IF Security documentation for integration !' Pass

% , IF ANY OF [Secure development lifecycle (SDL) , Test / type-approval reports available] !' Pass

% , ELSE !' Gap

Deterministic result

Pass

8.2 E27-LC2 — Vulnerability, patch & update management (UR E27 §3)

PASS

ASSESSMENT QUESTION

Has the organization implemented and evidenced "Vulnerability, patch & update management" as required by UR E27 §3?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1)

Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration

Established processes & evidence

Secure development lifecycle (SDL), Test / type-approval reports available, Security / hardening manual, Vulnerability / patch process

EVIDENCE

Evidenced: Signed / verified updates, Vulnerability / patch process.

Verification level

Declared

Needed to verify: Request a document or record substantiating the statement.

RULE LOGIC AND RESULT

% , IF Signed / verified updates !' Pass

% , IF ANY OF [Vulnerability / patch process , Defined support / EOL period] !' Partial

% , ELSE !' Gap

Deterministic result

Pass

8.3 E27-ID1 — Identification & authentication control (FR1) (UR E27 FR)

PASS

ASSESSMENT QUESTION

Has the organization implemented and evidenced "Identification & authentication control (FR1)" as required by UR E27 FR?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1)

Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration

EVIDENCE

Evidenced: Identification & authentication control (FR1).

Verification level

Declared

Needed to verify: Request a document or record substantiating the statement.

RULE LOGIC AND RESULT

% , IF Identification & authentication control (FR1) !' Pass

% , IF SOME OF [Identification & authentication control (FR1)] !' Partial

<pre>%, ELSE !' Gap</pre>	
Deterministic result	Pass

8.4 E27-PR1 — Use control & authorisation (FR2) (UR E27 FR2)

PASS

ASSESSMENT QUESTION

Has the organization implemented and evidenced "Use control & authorisation (FR2)" as required by UR E27 FR2?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1) Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration

EVIDENCE

Evidenced: Use control / authorisation (FR2).

Verification level	Declared
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Needed to verify: Request a document or record substantiating the statement.

RULE LOGIC AND RESULT

```
%, IF Use control / authorisation (FR2) !' Pass
%, IF SOME OF [ Use control / authorisation (FR2) ] !' Partial
%, ELSE !' Gap
```

Deterministic result	Pass
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8.5 E27-PR2 — System integrity & hardening (FR3) (UR E27 FR3)

PASS

ASSESSMENT QUESTION

Has the organization implemented and evidenced "System integrity & hardening (FR3)" as required by UR E27 FR3?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1) Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration

EVIDENCE

Evidenced: System integrity (FR3), Hardening & secure-by-default configuration.

Verification level	Declared
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Needed to verify: Request a document or record substantiating the statement.

RULE LOGIC AND RESULT

```
%, IF System integrity (FR3) !' Pass
```

<pre>%, IF ANY OF [Hardening & secure-by-default configuration] !' Partial %, ELSE !' Gap</pre>	
Deterministic result	Pass

8.6 E27-PR3 — Data confidentiality (FR4) (UR E27 FR4)

PASS

ASSESSMENT QUESTION

Has the organization implemented and evidenced "Data confidentiality (FR4)" as required by UR E27 FR4?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1)

Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration

EVIDENCE

Evidenced: Data confidentiality (FR4).

Verification level

Declared

Needed to verify: Request a document or record substantiating the statement.

RULE LOGIC AND RESULT

```
%, IF Data confidentiality (FR4) !' Pass
%, IF SOME OF [ Data confidentiality (FR4) ] !' Partial
%, ELSE !' Gap
```

Deterministic result

Pass

8.7 E27-PR4 — Restricted data flow (FR5) (UR E27 FR5)

FAIL

ASSESSMENT QUESTION

Has the organization implemented and evidenced "Restricted data flow (FR5)" as required by UR E27 FR5?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1)

Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration

EVIDENCE

No evidence provided in the intake data.

Verification level

Declared

Needed to verify: Request a document or record substantiating the statement.

RULE LOGIC AND RESULT

```
%, IF Restricted data flow (FR5) !' Pass
%, IF SOME OF [ Restricted data flow (FR5) ] !' Partial
%, ELSE !' Gap
```

Identified Gap: Not evidenced: Restricted data flow (FR5).

Deterministic result	Gap
Action	REC-01 — Implement / evidence: Restricted data flow (FR5).

8.8 E27-DE1 — Audit logging & timely response to events (FR6) (UR E27 FR6)

PASS

ASSESSMENT QUESTION

Has the organization implemented and evidenced "Audit logging & timely response to events (FR6)" as required by UR E27 FR6?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1) Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration

EVIDENCE

Evidenced: Audit logging & local recording, Timely response to events (FR6).

Verification level	Verified
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Needed to verify: Periodically confirm the verification record is still current.

RULE LOGIC AND RESULT

```
%, IF Audit logging & local recording !' Pass
%, IF ANY OF [ Timely response to events (FR6) ] !' Partial
%, ELSE !' Gap
```

Deterministic result	Pass
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8.9 E27-RS1 — Timely response to security events (FR6) (UR E27 FR6)

PASS

ASSESSMENT QUESTION

Has the organization implemented and evidenced "Timely response to security events (FR6)" as required by UR E27 FR6?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1) Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration

EVIDENCE

Evidenced: Timely response to events (FR6).

Verification level	Declared
Needed to verify: Request a document or record substantiating the statement.	
RULE LOGIC AND RESULT	
%, IF Timely response to events (FR6) !' Pass	
%, IF SOME OF [Timely response to events (FR6)] !' Partial	
%, ELSE !' Gap	
Deterministic result	Pass

8.10 E27-RC1 — Resource availability, robustness & recovery (FR7) (UR E27 FR7)

FAIL

ASSESSMENT QUESTION

Has the organization implemented and evidenced "Resource availability, robustness & recovery (FR7)" as required by UR E27 FR7?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1) Configuration & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration

EVIDENCE

No evidence provided in the intake data.

Verification level	Declared
Needed to verify: Request a document or record substantiating the statement.	
RULE LOGIC AND RESULT	
%, IF Resource availability / robustness (FR7) !' Pass	
%, IF ANY OF [Configuration backup & recovery] !' Partial	
%, ELSE !' Gap	
Identified Gap: Not evidenced: Resource availability / robustness (FR7).	
Deterministic result	Gap
Action	REC-02 — Implement / evidence: Resource availability / robustness (FR7).

Evidence Strength Overview

Source: Deterministic Assessment Engine

Informational overview of how far the evidence base has been substantiated. It does not affect the readiness score.

VERIFICATION LEVEL	
Verified	1 (10%)
Documented	1 (10%)
Declared	8 (80%)
No evidence	0 (0%)
EVIDENCE STRENGTH	
Very high	1 (10%)
High	1 (10%)
Medium	0 (0%)
Low	8 (80%)
REPORT METADATA	
Assessment ID	QA-1
Report Title	UR E27 Readiness Assessment
Report Version	1.0
Generated	12/08/2026, 17:38:58
Assessment Engine	det-1
AI Insight Engine	n/a

Working Papers & Assessment Traceability

Source: Deterministic Assessment Engine

For every requirement this working paper records exactly why it was assessed as Pass, Partial or Gap: each assessment question with the response given, the evidence recorded (type, name, source and strength), the deterministic rule applied, the generated risk and the AI sections that referenced the requirement. No assessment result is a black box.

1. E27-LC1 — Secure development lifecycle & evidence

Article: UR E27 §3 · Rule ID: E27-LC1-RULE

PASS

CONTROL OBJECTIVE

Secure development lifecycle & evidence (UR E27 §3)

ASSESSMENT QUESTION

Has the organization implemented and evidenced "Secure development lifecycle & evidence" as required by UR E27 §3?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1)	Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration
Established processes & evidence	Secure development lifecycle (SDL), Test / type-approval reports available, Security / hardening manual, Vulnerability / patch process
Comment	Evidenced: Security documentation for integration, Secure development lifecycle (SDL), Test / type-approval reports available.

EVIDENCE

Type	Document
Name	Secure development lifecycle & evidence — Document
Source	Documented evidence (referenced in intake)
Strength	High

Evidenced: Security documentation for integration, Secure development lifecycle (SDL), Test / type-approval reports available.

RESULT LOGIC

```
%, IF Security documentation for integration !' Pass
%, IF ANY OF [ Secure development lifecycle (SDL) , Test / type-approval reports av
%, ELSE !' Gap
```

RESULT

Deterministic Result	Pass
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RISK TRACEABILITY

No risk generated (requirement passed).

AI TRACEABILITY

Referenced by AI	No
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AUDIT TRAIL METADATA

Assessment ID	QA-1
Assessment Version	det-1
Requirement ID	E27-LC1
Rule ID	E27-LC1-RULE
Assessor	Inside the Box (rule-based assessment engine)
Source System	Inside the Box — Internal Audit & Compliance Readiness Platform
Timestamp	12/08/2026, 17:38:58

2. E27-LC2 — Vulnerability, patch & update management

Article: UR E27 §3 · Rule ID: E27-LC2-RULE

PASS

CONTROL OBJECTIVE

Vulnerability, patch & update management (UR E27 §3)

ASSESSMENT QUESTION

Has the organization implemented and evidenced "Vulnerability, patch & update management" as required by UR E27 §3?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1)	Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration
Established processes & evidence	Secure development lifecycle (SDL), Test / type-approval reports available, Security / hardening manual, Vulnerability / patch process
Comment	Evidenced: Signed / verified updates, Vulnerability / patch process.

EVIDENCE

Type	Statement
Name	Vulnerability, patch & update management — Statement
Source	Self-assessment (intake)
Strength	Low

Evidenced: Signed / verified updates, Vulnerability / patch process.

RESULT LOGIC

<pre>%, IF Signed / verified updates !' Pass %, IF ANY OF [Vulnerability / patch process , Defined support / EOL period] !' Part %, ELSE !' Gap</pre>	
RESULT	
Deterministic Result	Pass
RISK TRACEABILITY	
No risk generated (requirement passed).	
AI TRACEABILITY	
Referenced by AI	No
AUDIT TRAIL METADATA	
Assessment ID	QA-1
Assessment Version	det-1
Requirement ID	E27-LC2
Rule ID	E27-LC2-RULE
Assessor	Inside the Box (rule-based assessment engine)
Source System	Inside the Box — Internal Audit & Compliance Readiness Platform
Timestamp	12/08/2026, 17:38:58

3. E27-ID1 — Identification & authentication control (FR1)

Article: UR E27 FR · Rule ID: E27-ID1-RULE

PASS	
CONTROL OBJECTIVE	
Identification & authentication control (FR1) (UR E27 FR)	
ASSESSMENT QUESTION	
Has the organization implemented and evidenced "Identification & authentication control (FR1)" as required by UR E27 FR?	
ANSWER	
Security capabilities (UR E27 / IEC 62443-1-1)	Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration
Comment	Evidenced: Identification & authentication control (FR1).
EVIDENCE	
Type	Statement

Name	Identification & authentication control (FR1) — Statement
Source	Self-assessment (intake)
Strength	Low

Evidenced: Identification & authentication control (FR1).

RESULT LOGIC

%, IF Identification & authentication control (FR1) !' Pass
%, IF SOME OF [Identification & authentication control (FR1)] !' Partial
%, ELSE !' Gap

RESULT

Deterministic Result	Pass
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RISK TRACEABILITY

No risk generated (requirement passed).

AI TRACEABILITY

Referenced by AI	No
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AUDIT TRAIL METADATA

Assessment ID	QA-1
Assessment Version	det-1
Requirement ID	E27-ID1
Rule ID	E27-ID1-RULE
Assessor	Inside the Box (rule-based assessment engine)
Source System	Inside the Box — Internal Audit & Compliance Readiness Platform
Timestamp	12/08/2026, 17:38:58

4. E27-PR1 — Use control & authorisation (FR2)

Article: UR E27 FR2 · Rule ID: E27-PR1-RULE

PASS

CONTROL OBJECTIVE

Use control & authorisation (FR2) (UR E27 FR2)

ASSESSMENT QUESTION

Has the organization implemented and evidenced "Use control & authorisation (FR2)" as required by UR E27 FR2?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1)	Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration
Comment	Evidenced: Use control / authorisation (FR2).

EVIDENCE

Type	Statement
Name	Use control & authorisation (FR2) — Statement
Source	Self-assessment (intake)
Strength	Low

Evidenced: Use control / authorisation (FR2).

RESULT LOGIC

%, IF Use control / authorisation (FR2) !' Pass
%, IF SOME OF [Use control / authorisation (FR2)] !' Partial
%, ELSE !' Gap

RESULT

Deterministic Result	Pass
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RISK TRACEABILITY

No risk generated (requirement passed).

AI TRACEABILITY

Referenced by AI	No
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AUDIT TRAIL METADATA

Assessment ID	QA-1
Assessment Version	det-1
Requirement ID	E27-PR1
Rule ID	E27-PR1-RULE
Assessor	Inside the Box (rule-based assessment engine)
Source System	Inside the Box — Internal Audit & Compliance Readiness Platform
Timestamp	12/08/2026, 17:38:58

5. E27-PR2 — System integrity & hardening (FR3)

Article: UR E27 FR3 · Rule ID: E27-PR2-RULE

PASS

CONTROL OBJECTIVE

System integrity & hardening (FR3) (UR E27 FR3)

ASSESSMENT QUESTION

Has the organization implemented and evidenced "System integrity & hardening (FR3)" as required by UR E27 FR3?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1)

Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration

Comment

Evidenced: System integrity (FR3), Hardening & secure-by-default configuration.

EVIDENCE

Type

Statement

Name

System integrity & hardening (FR3) — Statement

Source

Self-assessment (intake)

Strength

Low

Evidenced: System integrity (FR3), Hardening & secure-by-default configuration.

RESULT LOGIC

% IF System integrity (FR3) !' Pass

% IF ANY OF [Hardening & secure-by-default configuration] !' Partial

% ELSE !' Gap

RESULT

Deterministic Result

Pass

RISK TRACEABILITY

No risk generated (requirement passed).

AI TRACEABILITY

Referenced by AI

No

AUDIT TRAIL METADATA

Assessment ID

QA-1

Assessment Version

det-1

Requirement ID

E27-PR2

Rule ID

E27-PR2-RULE

Assessor	Inside the Box (rule-based assessment engine)
Source System	Inside the Box — Internal Audit & Compliance Readiness Platform
Timestamp	12/08/2026, 17:38:58

6. E27-PR3 — Data confidentiality (FR4)

Article: UR E27 FR4 · Rule ID: E27-PR3-RULE

PASS

CONTROL OBJECTIVE

Data confidentiality (FR4) (UR E27 FR4)

ASSESSMENT QUESTION

Has the organization implemented and evidenced "Data confidentiality (FR4)" as required by UR E27 FR4?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1)

Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration

Comment

Evidenced: Data confidentiality (FR4).

EVIDENCE

Type

Statement

Name

Data confidentiality (FR4) — Statement

Source

Self-assessment (intake)

Strength

Low

Evidenced: Data confidentiality (FR4).

RESULT LOGIC

% IF Data confidentiality (FR4) !' Pass

% IF SOME OF [Data confidentiality (FR4)] !' Partial

% ELSE !' Gap

RESULT

Deterministic Result

Pass

RISK TRACEABILITY

No risk generated (requirement passed).

AI TRACEABILITY

Referenced by AI

No

AUDIT TRAIL METADATA

Assessment ID

QA-1

Assessment Version	det-1
Requirement ID	E27-PR3
Rule ID	E27-PR3-RULE
Assessor	Inside the Box (rule-based assessment engine)
Source System	Inside the Box — Internal Audit & Compliance Readiness Platform
Timestamp	12/08/2026, 17:38:58

7. E27-PR4 — Restricted data flow (FR5)

Article: UR E27 FR5 · Rule ID: E27-PR4-RULE

FAIL

CONTROL OBJECTIVE

Restricted data flow (FR5) (UR E27 FR5)

ASSESSMENT QUESTION

Has the organization implemented and evidenced "Restricted data flow (FR5)" as required by UR E27 FR5?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1)

Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration

Comment

No evidence provided in the intake data.

EVIDENCE

Type

Statement

Name

Restricted data flow (FR5) — Statement

Source

Self-assessment (intake)

Strength

Low

No evidence provided in the intake data.

RESULT LOGIC

%,' IF Restricted data flow (FR5) !' Pass

%,' IF SOME OF [Restricted data flow (FR5)] !' Partial

%,' ELSE !' Gap

RESULT

Deterministic Result

Gap

RISK TRACEABILITY

Generated Risk	R1
Risk Formula	Likelihood 3 × Impact 4 = 12 (Medium)
AI TRACEABILITY	
Referenced by AI	No
AUDIT TRAIL METADATA	
Assessment ID	QA-1
Assessment Version	det-1
Requirement ID	E27-PR4
Rule ID	E27-PR4-RULE
Assessor	Inside the Box (rule-based assessment engine)
Source System	Inside the Box — Internal Audit & Compliance Readiness Platform
Timestamp	12/08/2026, 17:38:58

8. E27-DE1 — Audit logging & timely response to events (FR6)

Article: UR E27 FR6 · Rule ID: E27-DE1-RULE

PASS

CONTROL OBJECTIVE

Audit logging & timely response to events (FR6) (UR E27 FR6)

ASSESSMENT QUESTION

Has the organization implemented and evidenced "Audit logging & timely response to events (FR6)" as required by UR E27 FR6?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1)	Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration
Comment	Evidenced: Audit logging & local recording, Timely response to events (FR6).

EVIDENCE

Type	Audit report
Name	Audit logging & timely response to events (FR6) — Audit report
Source	External audit/assurance report (referenced in intake)
Strength	Very high

Evidenced: Audit logging & local recording, Timely response to events (FR6).

RESULT LOGIC

%,	IF Audit logging & local recording !'	Pass
%,	IF ANY OF [Timely response to events (FR6)] !'	Partial
%,	ELSE !'	Gap

RESULT

Deterministic Result	Pass
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RISK TRACEABILITY

No risk generated (requirement passed).

AI TRACEABILITY

Referenced by AI	No
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AUDIT TRAIL METADATA

Assessment ID	QA-1
Assessment Version	det-1
Requirement ID	E27-DE1
Rule ID	E27-DE1-RULE
Assessor	Inside the Box (rule-based assessment engine)
Source System	Inside the Box — Internal Audit & Compliance Readiness Platform
Timestamp	12/08/2026, 17:38:58

9. E27-RS1 — Timely response to security events (FR6)

Article: UR E27 FR6 · Rule ID: E27-RS1-RULE

PASS

CONTROL OBJECTIVE

Timely response to security events (FR6) (UR E27 FR6)

ASSESSMENT QUESTION

Has the organization implemented and evidenced "Timely response to security events (FR6)" as required by UR E27 FR6?

ANSWER

Security capabilities (UR E27 / IEC 62443-1-1)	Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration
Comment	Evidenced: Timely response to events (FR6).

EVIDENCE

Type	Statement
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Name	Timely response to security events (FR6) — Statement
Source	Self-assessment (intake)
Strength	Low

Evidenced: Timely response to events (FR6).

RESULT LOGIC

%, IF Timely response to events (FR6) !' Pass
%, IF SOME OF [Timely response to events (FR6)] !' Partial
%, ELSE !' Gap

RESULT

Deterministic Result	Pass
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RISK TRACEABILITY

No risk generated (requirement passed).

AI TRACEABILITY

Referenced by AI	No
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AUDIT TRAIL METADATA

Assessment ID	QA-1
Assessment Version	det-1
Requirement ID	E27-RS1
Rule ID	E27-RS1-RULE
Assessor	Inside the Box (rule-based assessment engine)
Source System	Inside the Box — Internal Audit & Compliance Readiness Platform
Timestamp	12/08/2026, 17:38:58

10. E27-RC1 — Resource availability, robustness & recovery (FR7)

Article: UR E27 FR7 · Rule ID: E27-RC1-RULE

FAIL

CONTROL OBJECTIVE

Resource availability, robustness & recovery (FR7) (UR E27 FR7)

ASSESSMENT QUESTION

Has the organization implemented and evidenced "Resource availability, robustness & recovery (FR7)" as required by UR E27 FR7?

ANSWER

Security capabilities (UR E27 / IEC 62443-2-1)	Identification & authentication control (FR1), Use control / authorisation (FR2), System integrity (FR3), Data confidentiality (FR4), Timely response to events (FR6), Hardening & secure-by-default configuration, Audit logging & local recording, Signed / verified updates, Security documentation for integration
Comment	No evidence provided in the intake data.

EVIDENCE

Type	Statement
Name	Resource availability, robustness & recovery (FR7) — Statement
Source	Self-assessment (intake)
Strength	Low

No evidence provided in the intake data.

RESULT LOGIC

%, IF Resource availability / robustness (FR7) !' Pass
%, IF ANY OF [Configuration backup & recovery] !' Partial
%, ELSE !' Gap

RESULT

Deterministic Result	Gap
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RISK TRACEABILITY

Generated Risk	R2
Risk Formula	Likelihood 3 × Impact 5 = 15 (High)

AI TRACEABILITY

Referenced by AI	No
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AUDIT TRAIL METADATA

Assessment ID	QA-1
Assessment Version	det-1
Requirement ID	E27-RC1
Rule ID	E27-RC1-RULE
Assessor	Inside the Box (rule-based assessment engine)
Source System	Inside the Box — Internal Audit & Compliance Readiness Platform
Timestamp	12/08/2026, 17:38:58

Evidence Register (Appendix)

Source: Deterministic Assessment Engine

Every evidence item mapped to its requirement, type, strength and assessment contribution.

ID	Req.	Type	Strength	Result
E-001	E27-LC1	Document	High	Pass
E-002	E27-LC2	Statement	Low	Pass
E-003	E27-ID1	Statement	Low	Pass
E-004	E27-PR1	Statement	Low	Pass
E-005	E27-PR2	Statement	Low	Pass
E-006	E27-PR3	Statement	Low	Pass
E-007	E27-PR4	Statement	Low	Gap
E-008	E27-DE1	Audit report	Very high	Pass
E-009	E27-RS1	Statement	Low	Pass
E-010	E27-RC1	Statement	Low	Gap

E-001 — E27-LC1

Requirement	Secure development lifecycle & evidence
Type	Document
Strength	High
Used For	Secure development lifecycle & evidence
Result Contribution	Pass

DESCRIPTION

Evidenced: Security documentation for integration, Secure development lifecycle (SDL), Test / type-approval reports available.

E-002 — E27-LC2

Requirement	Vulnerability, patch & update management
Type	Statement
Strength	Low
Used For	Vulnerability, patch & update management
Result Contribution	Pass

DESCRIPTION

Evidenced: Signed / verified updates, Vulnerability / patch process.

E-003 — E27-ID1

Requirement	Identification & authentication control (FR1)
Type	Statement
Strength	Low
Used For	Identification & authentication control (FR1)
Result Contribution	Pass

DESCRIPTION

Evidenced: Identification & authentication control (FR1).

E-004 — E27-PR1

Requirement	Use control & authorisation (FR2)
Type	Statement
Strength	Low
Used For	Use control & authorisation (FR2)
Result Contribution	Pass

DESCRIPTION

Evidenced: Use control / authorisation (FR2).

E-005 — E27-PR2

Requirement	System integrity & hardening (FR3)
Type	Statement
Strength	Low
Used For	System integrity & hardening (FR3)
Result Contribution	Pass

DESCRIPTION

Evidenced: System integrity (FR3), Hardening & secure-by-default configuration.

E-006 — E27-PR3

Requirement	Data confidentiality (FR4)
Type	Statement

Strength	Low
Used For	Data confidentiality (FR4)
Result Contribution	Pass

DESCRIPTION

Evidenced: Data confidentiality (FR4).

E-007 — E27-PR4

Requirement	Restricted data flow (FR5)
Type	Statement
Strength	Low
Used For	Restricted data flow (FR5)
Result Contribution	Gap

DESCRIPTION

No evidence provided in the intake data.

E-008 — E27-DE1

Requirement	Audit logging & timely response to events (FR6)
Type	Audit report
Strength	Very high
Used For	Audit logging & timely response to events (FR6)
Result Contribution	Pass

DESCRIPTION

Evidenced: Audit logging & local recording, Timely response to events (FR6).

E-009 — E27-RS1

Requirement	Timely response to security events (FR6)
Type	Statement
Strength	Low
Used For	Timely response to security events (FR6)
Result Contribution	Pass

DESCRIPTION

Evidenced: Timely response to events (FR6).

E-010 — E27-RC1

Requirement	Resource availability, robustness & recovery (FR7)
Type	Statement
Strength	Low
Used For	Resource availability, robustness & recovery (FR7)
Result Contribution	Gap

DESCRIPTION

No evidence provided in the intake data.

Appendix B Scoring Methodology

Source: Deterministic Assessment Engine

This appendix documents exactly how every score and readiness percentage in this report is calculated, so each figure is fully reproducible from the recorded answers.

Control scoring

- %, Each control is scored deterministically: Pass = 100, Partial = 50, Gap = 0.
- %, A control is Pass only when all required evidence tokens are present, Partial when some are present, Gap when none are.

Readiness score

- %, Readiness % = weighted average of control scores, where each control carries its defined weight (default 1).
- %, Formula: $\text{sum}(\text{controlScore} \times \text{weight}) \div \text{sum}(\text{weight})$, rounded to the nearest integer.
- %, Bands: Strong "e 80 %, Substantial "e 60 %, Developing "e 35 %, Limited < 35 %.

Audit readiness dimensions

Documentation	Controls backed by documented evidence (policy/procedure/document/audit report) ÷ total controls.
Operational	The weighted compliance score — effectiveness of implemented controls.
Governance	Average score of governance-related categories; overall score when no such category exists.
Evidence	Coverage weighted by evidence strength (Low 25, Medium 50, High 75, Very high 100) ÷ total controls.

Overall audit readiness = mean of the four dimension percentages.

Risk scoring

- %, One risk is derived per non-passing control. Risk score = Likelihood × Impact (1–5 scale).
- %, Default likelihood: 4 for a Gap, 3 for a Partial; impact defaults to 3 and can be tuned per control.
- %, Ratings: Critical "e 20, High "e 13, Medium "e 6, Low < 6.